

Two Businesses Under One Roof:

An End-to-End Analysis of NYC Yellow Taxi Trips (2024)



A portfolio project in Python, SQL and Excel

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Data source: NYC Taxi & Limousine Commission (TLC) Trip Record Data

Observations analysed: 35,636,331 cleaned trips

Tools: **Python** (pandas, pyarrow, geopandas, matplotlib) · **MySQL** · **Microsoft Excel**

Preface

This project began as an exercise in working through a complete analytical workflow on data that had not been prepared for the purpose. The trip records published by the New York City Taxi & Limousine Commission seemed a reasonable place to start: they are freely available, they contain geodata, they cover an entire year, and they arrive with all the difficulties of genuine administrative data, an unfamiliar file format, roughly forty-one million rows, a share of corrupted observations, and categorical variables encoded as bare integers that mean nothing without the accompanying documentation.

Working with a public dataset also makes it possible to check one's own results against outside reference points. Where an official TLC figure exists it is compared directly; where only figures from other independent analyses of the same data exist, the comparison is weaker and is treated as such. This distinction turns out to matter, and Section 6 examines it honestly: reproducing a number that others also computed from the raw records confirms consistency, not correctness.

The tools were divided according to what each does well. *Python handles the file conversion and the automation around it and the geospatial mapping of the taxi zones. SQL carries the computational weight, which on a table of this size is considerable. Excel takes the finished results and renders them legible.*

NOTE: originally multiple files were separately created; in order to not force the reader to perform countless individual downloads, they have ultimately been aggregated all together, separated by:

“#####”

1. Motivation and Research Questions

Urban mobility data is among the richest and least exploited sources of behavioural evidence available to analysts. Every taxi trip is a small, timestamped, priced decision: someone decided to travel, from somewhere, at some hour, and paid for it. Aggregated across tens of millions of such decisions, patterns emerge that no survey could reconstruct, patterns about how a city works, when it works, and who pays for it.

The New York yellow taxi is a particularly interesting object of study because it operates under a peculiar regulatory settlement. Yellow medallion taxis hold the exclusive right to accept street hails across all five boroughs, yet they face intense competition from ride-hailing platforms and, since 2013, from the green “boro taxis” introduced specifically because yellow cabs were failing to serve the outer boroughs. Whether that failure is visible in the data is an empirical question, and one this project answers.

The analysis is organised around four questions:

Q1. How does demand vary across time, across the year and across the hours of the day and does the composition of that demand change with it?

Q2. What does the payment data reveal about rider behaviour, and where does the recording mechanism itself distort what we observe?

Q3. How is taxi activity distributed geographically, and does the yellow cab genuinely serve New York City or only a fraction of it?

Q4. How far can the figures I compute independently be checked against external sources, and what does such a check actually establish?

2. Data and Sources

The TLC publishes trip records monthly, with a roughly two-month lag to allow for complete vendor submissions. Each row represents a single completed trip and is generated by the Technology Service Providers that operate the meters, not by the TLC itself, which means the agency explicitly declines to guarantee accuracy or completeness. This caveat is relevant since it has direct consequences for the cleaning stage described below.

I downloaded the twelve monthly files covering January to December 2024 (**all links to the data are provided in the references section**). The files are distributed in PARQUET, a compressed columnar format that cannot be opened by Excel and cannot be loaded directly into MySQL. This is the first obstacle the project has to solve, and it is the reason Python enters the pipeline at all.

Alongside the trip records, two supporting documents are essential. The Taxi Zone Lookup Table maps the 265 numeric zone identifiers used in the trip files to borough and neighbourhood names, without it, a pickup location is simply the integer 132, which is meaningless. The Data Dictionary defines every field and, critically, decodes the categorical variables.

2.1 Variable description

The nineteen fields of the yellow taxi records, with the coding required to interpret them:

Variable	Type	Key values / description
VendorID	categorical	Technology provider: 1 = Creative Mobile Tech.; 2 = VeriFone/Curb
tpcp_pickup_datetime	datetime	Moment the meter was engaged
tpcp_dropoff_datetime	datetime	Moment the meter was disengaged
passenger_count	numeric	Passengers in the vehicle — entered manually by the driver
trip_distance	numeric	Elapsed distance in miles, reported by the taximeter
RatecodeID	categorical	1 = Standard; 2 = JFK (flat fare); 3 = Newark; 4 = Nassau/Westchester; 5 = Negotiated; 6 = Group ride
store_and_fwd_flag	binary	Y = record held in vehicle before transmission (no connection); N otherwise
PULocationID	categorical	TLC Taxi Zone where the meter was engaged (1–265)
DOLocationID	categorical	TLC Taxi Zone where the meter was disengaged (1–265)
payment_type	categorical	1 = Credit card; 2 = Cash; 3 = No charge; 4 = Dispute; 5 = Unknown; 6 = Voided
fare_amount	numeric	Time-and-distance fare computed by the meter
tip_amount	numeric	Tip — populated for card payments only; cash tips are not captured
tolls_amount	numeric	Total tolls paid during the trip
total_amount	numeric	Total charged to the passenger, excluding cash tips
extra, mta_tax, improvement_surcharge, congestion_surcharge, airport_fee	numeric	Itemised surcharges and taxes; airport_fee applies to JFK and LaGuardia pickups

Two features of this table deserve emphasis, because they shape every result that follows. First, *tip_amount* is populated only for electronic payments; a cash tip is invisible to the system. Second, *fare_amount* is what the meter computed, not what was collected; a trip can carry a positive fare and still be recorded as “no charge”. Confusing the two would produce badly wrong revenue estimates.

3. Methodology: The Pipeline

The architecture of the project follows the division of labour a professional analyst would adopt: Python automates the ingestion and handles geospatial visualization, SQL performs the heavy computation, and Excel handles the tabular presentation.

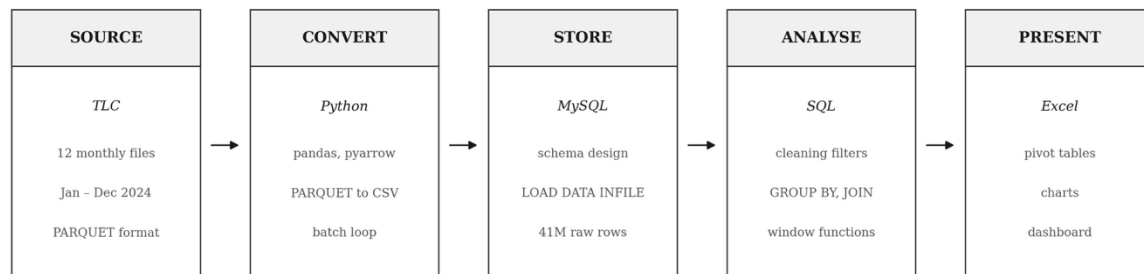


Figure 1 — Project pipeline: raw PARQUET files are converted to CSV in Python, bulk-loaded into MySQL, cleaned into a separate table, analysed with SQL, and exported to Excel for visualisation.

3.1 Python: format conversion

The twelve PARQUET files total roughly 700 MB compressed and cannot be read by either of the downstream tools. A short script iterates over the directory, reads each file with pandas (backed by the pyarrow engine) and writes the corresponding CSV. Automating the loop rather than converting files by hand matters for reproducibility: the script can be re-run from scratch on a new year of data with a single path change.

It is worth being explicit about why this step exists at all, because it illustrates a judgement call. **I chose the CSV route because the stated objective of the project is to use MySQL**, and because bulk-loading flat files into a relational database is the more common workflow in production environments.

3.2 SQL: loading and schema

The database schema declares the nineteen columns with their appropriate types: DATETIME for the two timestamps, INT for the categorical codes, DOUBLE for the monetary and distance fields. All twelve monthly files are then bulk-loaded into a single table using LOAD DATA LOCAL INFILE, which reads a flat file directly into storage and is orders of magnitude faster than row-by-row insertion. Loading the full year takes approximately three minutes and produces a table of roughly 41 million rows (considering only the observations of 2024; I choose this year because is complete and sound).

One configuration obstacle is worth recording: the client's default query timeout of thirty seconds is far too short for operations on tens of millions of rows and must be raised.

3.3 SQL: data cleaning

The TLC's warning about accuracy is fully justified. The raw data contains trips with pickup timestamps outside the year entirely, negative fares, zero-mile journeys, and trips carrying zero passengers. These are not edge cases to be reasoned about; they are corrupted records, and including them would contaminate every average in the analysis.

The cleaning strategy deserves comment because it reflects a principle rather than a convenience. Rather than deleting rows from the loaded table, I created a separate clean table using CREATE TABLE ... AS SELECT, applying the filters at construction. The raw table is left untouched and permanently available.

This costs disk space but buys two things: the cleaning is fully reversible, and the raw data remains as an auditable reference. **Destructive operations on a source table are irreversible in SQL**, there is no undo, and the discipline of never modifying the original is standard practice for good reason.

The filters applied were: pickup timestamp within calendar year 2024; fare amount non-negative; trip distance strictly positive; passenger count strictly positive.

Stage	Rows	Share
Raw records loaded (12 months)	41,169,720	100%
Records removed by cleaning filters	5,533,389	≈ 13%
Clean analytical dataset	35,636,331	≈ 87%

Discarding thirteen per cent of a dataset is not a trivial intervention and should be acknowledged as such. It is possible that some removed records were valid trips with a single corrupted field, and a more sophisticated approach would attempt field-level repair rather than row-level deletion.

3.4 Relational structure

The trip records store pickup and dropoff locations as integers. The Taxi Zone Lookup table, loaded as a second table, maps those integers to borough and zone names: PULocationID and DOLocationID in the trip table both correspond to LocationID in the lookup table. Every geographic result in Section 5 is obtained by joining the two on this key. The foreign key constraint was not declared on the trip table. Enforcing it on thirty-five million rows would be costly and would in any case fail on identifiers 264 and 265, which encode "Unknown" and "Outside NYC" rather than real zones and therefore have no valid parent row. The relationship is respected in every query; it is simply not registered with the engine.

4. Results: Temporal and Behavioural Patterns

4.1 Seasonality

Monthly volumes describe an annual cycle with a clear structure. Trip counts are flat through the winter, expand sharply in March (+11.6% on February), and then decline steadily through the summer. July registers the largest contraction of the year at −10.8%, and August is the weakest month in absolute terms with 2.60 million trips. September then produces the strongest expansion of the year, +16.2%, and October becomes the annual peak at 3.30 million trips.

The interpretation is more interesting than it first appears. If yellow taxi demand were driven primarily by tourism, the peak would fall in summer, when visitors arrive. It does not; summer is the trough. The cycle instead tracks the working calendar: activity collapses when New Yorkers leave the city in July and August, and rebounds violently when they return in September. The yellow taxi is a commuting and business instrument, not a tourist one. Demand also tracks the working week, climbing from a Sunday low of 4.40 million trips to a Thursday peak of 5.59 million (+27%).

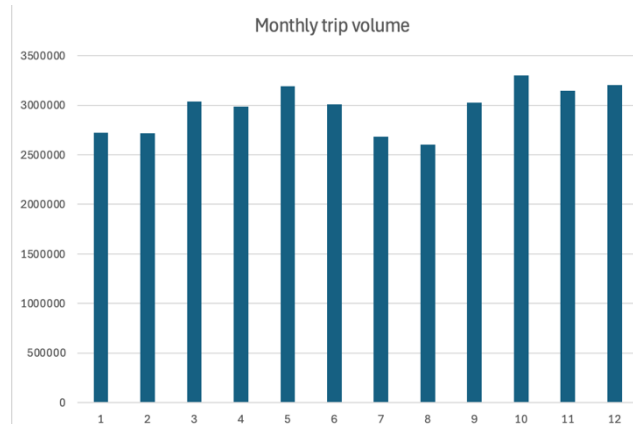


Figure 2 — Monthly trip volume, 2024. The summer contraction (July −10.8%) and the September rebound (+16.2%) trace the working calendar rather than the tourist season.

4.2 The intraday paradox

Hourly aggregation produces the single most informative result of the project, and it emerges only when two metrics are read together.

Trip volume follows the expected urban profile: a minimum of 166,992 trips at 4 a.m., rising steadily through the day to a peak of 2,557,789 at 6 p.m., a fifteen-fold swing between the quietest and busiest hour. Read alone, this tells us very little we could not have guessed.

Average trip distance, however, moves in the opposite direction. At 6 p.m., when demand peaks, the average trip is 3.08 miles; the shortest of the entire day. At 5 a.m., when demand is near its minimum, the average trip is 8.48 miles, nearly three times longer. The progression is abrupt: 3.39 miles at 3 a.m., 4.96 at 4 a.m., 8.48 at 5 a.m., then back down to 5.35 by 6 a.m.

The two curves are not describing the same phenomenon. The evening peak is intra-urban movement; short hops across Midtown in congested traffic, high frequency, low value per trip. The dawn spike is airport traffic: passengers leaving for early flights, few in number but travelling a long way. The yellow taxi is performing two entirely distinct functions, and they are separated in time. The aggregate hides this completely; only the joint reading reveals it.

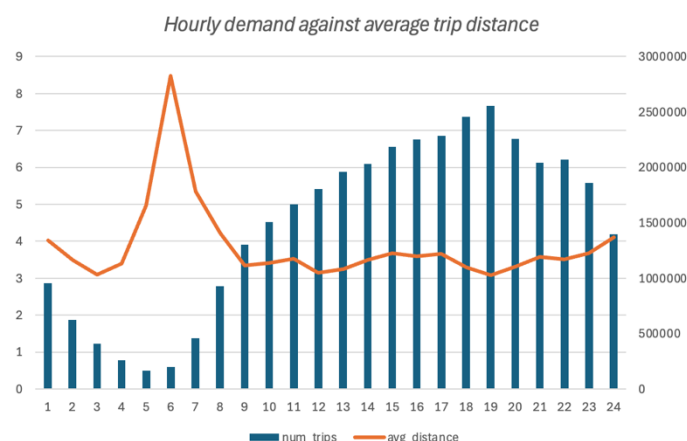


Figure 3 — Hourly demand against average trip distance (secondary axis). The curves move inversely: the busiest hour carries the shortest trips, while the dawn trough carries the longest. The 5 a.m. spike is the statistical signature of airport departures.

4.3 Payment and the visibility of tips

Card payments account for 83.8% of trips and cash for 14.65%, with disputes and no-charge trips making up the residual 1.5%. The behavioural finding, however, lies in the tip column.

Payment type	Trips	Avg fare	Avg tip	Avg total	Tip %
Credit card	29,875,818	\$19.75	\$4.36	\$29.76	25.1%
Cash	5,219,199	\$19.65	\$0.00	\$24.97	0.0%
Dispute	384,317	\$22.30	\$0.01	\$27.95	0.1%
No charge	149,721	\$22.15	\$0.01	\$27.48	0.1%

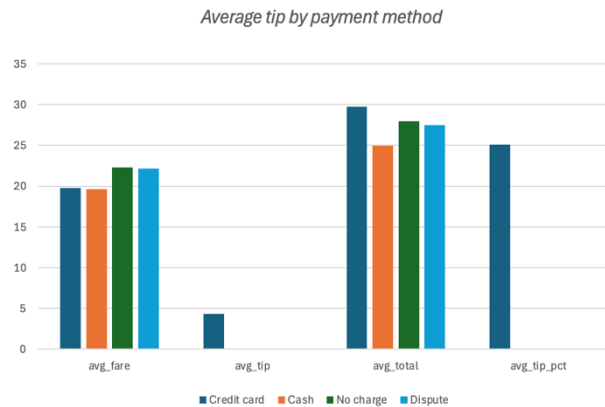
Card payers tip an average of \$4.36, or 25.1% of the fare. Cash payers tip, apparently, nothing at all. The naive reading (that people carrying cash are ungenerous) is false, and the table itself contains the refutation: the average fare is essentially identical across the two groups (\$19.75 versus \$19.65). These are not different kinds of passengers taking different kinds of trips. They are the same passengers, and the difference lies entirely in the recording mechanism. A tip handed to a driver in cash never passes through the electronic system and is therefore absent from the data.

This has a direct methodological consequence. Computing an average tip across the whole dataset (treating the cash zeros as observed zeros rather than missing values) would return roughly 21%, a figure that is simply wrong. It mixes measured values with unmeasured ones. The correct statement is

that the average tip among card payers is 25.1%, and that cash tipping behaviour is unobservable in this dataset.

The 25.1% figure is itself worth a remark. It is high by international standards but consistent with the American norm, and it is almost certainly shaped by the payment terminal itself, which presents preset options typically at 20%, 25% and 30%. The average settles close to the middle default. The design of the interface is visible in the distribution of the outcome.

The dispute and no-charge categories carry a further lesson. Both show positive average fares, \$22.30



and \$22.15, which is at first sight contradictory: how can a trip that generated no charge have a fare? The answer is that *fare_amount* records what the meter computed, not what was collected. The trip happened; the payment did not. Note also that both anomalous categories carry fares roughly 13% above the ordinary trip. This is not random: disputes arise disproportionately on longer, more expensive journeys, where there is both more that can go wrong and more financial incentive to complain.

Figure 4 — Average tip and fare by payment method. The apparent absence of cash tipping is an artefact of electronic recording, not a behavioural finding: average fares are near-identical across the two groups.

5. Results: The Geography of the Yellow Taxi

Resolving the numeric zone identifiers into place names, through the join described in Section 3.4, transforms the geographic questions from unanswerable to trivial.

5.1 Borough concentration

Manhattan accounts for 88.86% of all pickups. Queens accounts for 9.79%. Every other borough combined, Brooklyn, the Bronx, Staten Island, accounts for less than 1.1%. Staten Island, a borough of some half a million residents, generated 684 yellow taxi pickups in the entire year: roughly two per day, out of thirty-five million.

The Queens figure requires care, and it is here that the geographic and the economic analysis converge. Queens appears, at first glance, to be a meaningfully served borough. It is not. JFK and LaGuardia are both located in Queens, and between them they generated 3.10 million pickups. Total Queens pickups were 3.49 million. Airport traffic therefore constitutes approximately 89% of all yellow taxi activity in Queens; strip it out, and the borough's genuine residential share falls below 1.1%, in line with Brooklyn.

The conclusion is unambiguous and can be stated flatly: the yellow taxi is a Manhattan service with an airport shuttle attached. The outer boroughs, where the majority of New Yorkers actually live, are effectively unserved. This is precisely the market failure that motivated the introduction of the green boro taxi programme in 2013, and the 2024 data shows it persisting.

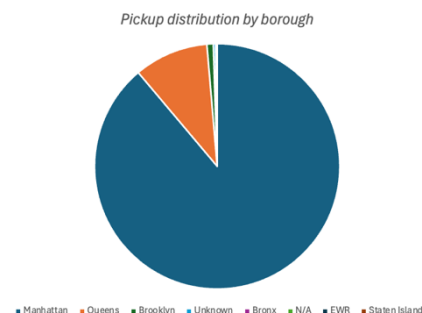


Figure 5 — Pickup distribution by borough. Manhattan's 88.86% share renders the chart almost monochromatic — which is itself the finding. Queens' 9.79% is overwhelmingly airport traffic.

NYC Taxi Zones by Borough

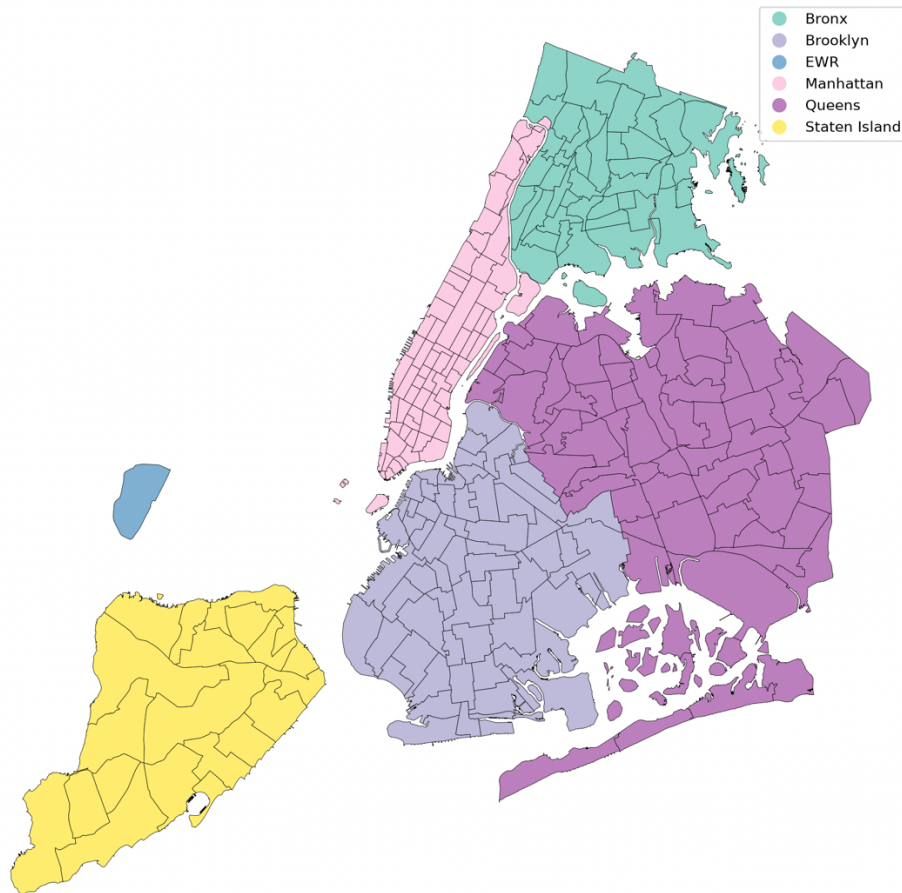
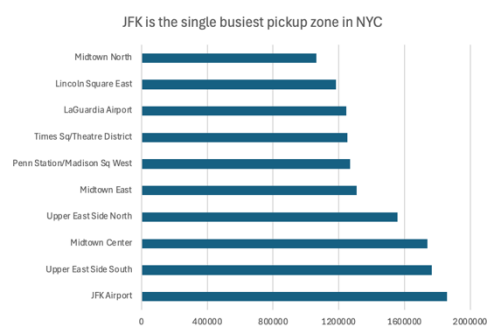


Figure 6 - NYC Taxi Zones by Borough. A spatial overview that highlights the stark contrast between the city's vast landmass and the narrow, Manhattan-centric footprint of yellow cabs. See Geodata in the links.

5.2 Zone-level ranking

The busiest single pickup zone in New York City is not in Manhattan. It is JFK Airport, with 1,857,707 pickups, ahead of every Manhattan neighbourhood. LaGuardia ranks eighth with 1,245,714.



The Manhattan zones that fill the intervening ranks form a coherent portrait: Upper East Side South and North (affluent residential), Midtown Center, Midtown East and Midtown North (the office core), Penn Station (the rail interchange), Times Square (tourism and theatre), and Lincoln Square. Read together, the top ten describe three distinct customer bases, the wealthy residents of the Upper East Side, the Midtown workforce, and travellers passing through the airports and Penn Station.

Figure 7 — Top ten pickup zones. JFK Airport outranks every Manhattan neighbourhood, and LaGuardia places eighth — the two airports together generate more pickups than any single district of the city.

5.3 The economics of airport trips

Origin	Trips	Avg distance	Avg fare	Avg total
JFK Airport	1,857,707	16.02 mi	\$64.27	\$82.77
LaGuardia Airport	1,245,714	9.74 mi	\$44.17	\$68.55
City average (all trips)	35,636,331	≈ 3.30 mi	≈ \$19.75	≈ \$29.00

A JFK trip travels nearly five times the distance of an average city trip and generates over three times the fare. The gap between JFK and LaGuardia is explained by geography and by regulation: JFK is further from Manhattan, and it operates under a flat fare (RatecodeID 2) rather than a running meter, which both raises and standardises the price.

A rough revenue decomposition makes the strategic point. The airports account for approximately 8.7% of trips but, at an average total of roughly \$77, contribute in the order of \$240 million against an estimated annual total near \$1.07 billion, approximately 22% of revenue. This asymmetry is the reason the dawn distance spike identified in Section 4.2 matters commercially and not merely as a curiosity. The airport segment is small in volume, disproportionate in value, and temporally concentrated.

6. External Checks and Their Limits

Every figure above was computed independently from raw records. It is worth being precise about what "validation" can and cannot mean here, because the honest answer is more limited than it first appears.

The New York City Taxi & Limousine Commission publishes very little in the way of aggregate behavioural statistics. Its 2020 Factbook, drawing on 2018–2019 data, is the one official source that offers a directly comparable figure: it reports that 97% of medallion pickups occurred in Manhattan. The other benchmarks widely quoted for this dataset (the single-passenger share, the cash share, the busiest weekday, the airport share) are not, as far as I can establish, published by the TLC in any report. They circulate through independent analyses of the same raw trip records. Reproducing them therefore confirms that I read the data the same way others do; it does not constitute validation against an external ground truth, because the "benchmark" and my figure are computed from the same source.

Metric	This analysis	Reference	Nature of comparison
Manhattan share of pickups	88.86%	97% (TLC Factbook, 2018–19)	Official but different years
Single passenger trips	78.14%	78% (independent analysis)	Same source agreement
Busiest weekday	Thursday	Thursday (independent analysis)	Same source agreement
Quietest weekday	Sunday, then Monday	Monday (independent analysis)	Same source agreement
Airport share of trips	8.7%	≈ 8% (independent analysis)	Same source agreement
Cash payments	14.66%	≈ 10% (independent analysis)	Divergent

Only the first row is a genuine external check, and it does not match: my 2024 Manhattan share is 88.86% against the Factbook's 97% for 2018–19. That gap is not an error but a finding; the Manhattan concentration has fallen by roughly eight points in five years, consistent with the airport segment growing as a share of a shrinking whole. The remaining four agreements show only that my pipeline processes the data conventionally; they are reassuring about internal consistency, not proof of correctness. The cash figure is examined separately below.

The cash payment figure does not reconcile, and it would be dishonest to present it as though it did. My estimate of 14.66% exceeds the commonly cited 10%. Two explanations are plausible: the cleaning filters may remove card and cash trips at different rates, or the official statistic may be computed over a different base.

Both can be tested, and both fail. On the raw table, before any filtering, the cash share is already 13.46%; the filters nudge it up to 14.66% by removing proportionally more card trips, but the divergence was there from the start. As for the base, the raw data contains 4,091,232 Flex Fare trips (payment_type 0), 9.94% of all records, which are neither card nor cash, but removing them from the denominator returns

14.94%, higher rather than lower. In retrospect this is arithmetically inevitable: shrinking a denominator can only raise a share.

The discrepancy therefore remains unexplained. The $\approx 10\%$ figure itself comes from independent analyses rather than any TLC publication, and the likeliest candidates are the reference period behind it (the Flex Fare payment type was still expanding through 2024–25) or a definitional difference I have not identified.

7. Conclusion and Reflection

The four results of this project converge on a single proposition. The New York yellow taxi is not one business but two, operating under the same medallion and largely invisible to each other in the aggregate statistics.

The first is a Manhattan commuting service: extremely high volume, very short trips, concentrated in the evening peak, priced at roughly twenty dollars, and geographically confined to less than a fifth of the city's landmass. The second is an airport shuttle: low volume, very long trips, concentrated at dawn, priced at three to four times the city average, operating out of Queens, and contributing a share of revenue more than double its share of trips. The two have inverted hourly profiles, distinct geographies, and different economics. Every one of the analyses above is a different projection of this same underlying structure.

The regulatory implication is uncomfortable. Yellow taxis hold the exclusive privilege of accepting street hails across all five boroughs, yet in practice they serve one borough and two airports. Brooklyn, home to more people than Manhattan, generated 0.84% of pickups. This is not a subtle finding; it is a system-level failure of coverage, and it validates the reasoning behind the boro taxi programme.

7.1 Limitations

Several limitations should be stated plainly, since a portfolio project that presents itself as flawless is not credible.

The month-on-month growth figures in Section 4.1 are not length-adjusted. February has 29 days and January 31, and part of the observed variation is calendar arithmetic rather than behaviour. A more rigorous treatment would normalise to average trips per day, and this is the single most obvious improvement available.

The cleaning filters are blunt. Thirteen per cent of records were discarded on row-level rules, and some of those rows may have been valid trips with one corrupted field. Field-level imputation would preserve more data at the cost of considerable additional complexity.

The revenue decomposition in Section 5.3 rests on averages rather than on a summed total, and should be read as an order-of-magnitude estimate rather than an accounting figure. The distributions of fares are heavily right-skewed, and a mean-based extrapolation across thirty-five million trips accumulates error.

Finally, the analysis is entirely descriptive. It establishes that the dawn distance spike coincides with airport traffic; it does not prove causation, though the mechanism is not seriously in doubt. Nothing here identifies a causal effect, and no claim in this report should be read as though it did.

7.2 Extensions

The most immediately valuable extension would be to analyse dropoff zones alongside pickups. This project examines only where trips begin; the flow structure (which pairs of zones are connected, and how asymmetrically) would require a double join against the lookup table and would likely reveal directional patterns invisible in the pickup data alone. Average speed by hour and zone, computed from distance and elapsed time, would map congestion. And a comparison with the green boro taxi records, published in the same format, would allow a direct test of whether the coverage gap identified here is genuinely being filled.

7.3 Tools and competences demonstrated

Python: pandas and pyarrow for PARQUET-to-CSV conversion; automated iteration over a file directory; defensive handling of pre-existing outputs; geopandas and matplotlib for parsing spatial data (Shapefiles) and rendering categorical maps.

SQL (MySQL): schema design and type declaration; bulk loading via LOAD DATA LOCAL INFILE; server and client configuration; non-destructive cleaning through CREATE TABLE AS SELECT; indexing for performance; aggregation with GROUP BY; window functions (OVER, LAG); common table expressions; multi-table joins; relational modelling.

Excel: import and type coercion of externally generated data; dual-axis composite charts; pie, bar and column charts; conditional formatting; dashboard assembly.

Analytical judgement: distinguishing behavioural findings from recording artefacts; identifying missing data misencoded as zero; testing results against external reference points and reporting honestly where no true external benchmark exists; surfacing a divergent figure rather than suppressing it.

Sources

NYC Taxi & Limousine Commission — TLC Trip Record Data (raw PARQUET files, Taxi Zone Lookup Table):

<https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

NYC TLC — Data Dictionary, Yellow Taxi Trip Records (variable definitions and categorical codes):

https://www.nyc.gov/assets/tlc/downloads/pdf/data_dictionary_trip_records_yellow.pdf

NYC TLC — Trip Record User Guide:

https://www.nyc.gov/assets/tlc/downloads/pdf/trip_record_user_guide.pdf

NYC TLC — Industry Reports and Factbook (2020 Factbook, source of the Manhattan pickup figure)

<https://www.nyc.gov/site/tlc/about/industry-reports.page>

NYC TLC — 2024 Annual Report:

https://www.nyc.gov/assets/tlc/downloads/pdf/annual_report_2024.pdf

ROW ZERO — NYC Taxi Data summary (independent analysis of 2024–25 TLC trip)

<https://rowzero.com/datasets/nyc-taxi-data>